02/11/26

Simulated Annealing

```
current ← initial state
T ← a large positive value
while T > 0 do
    next ← a random neighbour of current
    ΔE ← current cost - next cost
    if ΔE < 0 then
        current ← next
    else
        current ← next with probability P e^(ΔE/T)
    end if
    decrease T
end while
return current.
```

Output:

```
Enter the row position of each queen (1-8):
Queen 1 : 3
Queen 2 : 1
Queen 3 : 2
Queen 4 : 0
Queen 5 : 7
Queen 6 : 5
Queen 7 : 6
Queen 8 : 4
```

Final Board:

```
- - - - - - Q -
- - - Q - - - -
Q - - - - - - -
- - - - Q - - -
- - - - - - - Q
- Q - - - - - -
- - - - - Q - -
- - Q - - - - -
```

Queen Placements:
[3, 6, 8, 2, 4, 1, 7, 5]

Conflicts: 0

Solution Found.